

QIMIN FENG

Ph.D. Candidate in Mechanical Engineering | Robotics Control · Multimodal Perception · Systems Integration

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TARGET ROLE: Robotics R&D Engineer | Embedded Control, Multisensor Perception & Mechatronic Integration

SUMMARY

Mechanical Engineering Ph.D. candidate specializing in bio-inspired underwater and soft robotics. Builds end-to-end robotic systems—from architecture, mechanisms, and actuators to embedded control, multisensor perception, integration, and underwater validation—delivering **controllable, measurable, experimentally verified hardware**.

EDUCATION

Iowa State University | Ph.D. Candidate, Mechanical Engineering

Expected Dec. 2027

GPA: 3.97 / 4.00 | Research: Bio-inspired robotics, underwater locomotion, fluid-structure interaction, and flow sensing

Zhejiang University | B.Eng., Agricultural and Biological Systems Engineering

June 2022

CORE SKILLS

Controls & Embedded Systems: Python, Embedded C / C++, Raspberry Pi 5, ESP32-C6, Teensy, Arduino, Linux / systemd; multirate real-time control, open-loop actuation, PID feedback control, trajectory generation, safety state machines, watchdogs, and fail-safes

Communication & System Integration: CAN / SocketCAN / MIT Motor Protocol, ESP-NOW, I2C, PWM, serial, UDP / WebSocket; BLDC motors, servos, VCAs, IMUs, pressure / depth / Hall sensors, and stereo cameras

Mechanical Design & Technical Communication: SolidWorks (system architecture, parts, assemblies, and experimental fixtures), KeyShot, Adobe Illustrator, and 3D printing

Multimodal Perception & Data: multisensor acquisition, synchronization, calibration, and fusion; OpenCV stereo calibration, StereoSGBM depth estimation, PyTorch, attention networks, uncertainty estimation, sparse-to-full-field reconstruction, and NumPy / SciPy / MATLAB

Experiments & Testing: PIV (LaVision), ATI Mini-40, NI DAQ, Simulink Real-Time, CPS four-axis platform, and recirculating water-tunnel testing

ROBOTICS EXPERIENCE

Iowa State University · Zhong Lab | Graduate Research Assistant

Jan. 2023–Present

MARIS Manta-Inspired Underwater Robot | Control, Teleoperation & Stereo Perception

2026–Present

Raspberry Pi 5 · Python · SocketCAN · OpenCV · PySide6 / QML · Steam Deck

- Designed the MARIS system architecture and body in SolidWorks; integrated dual motors, dual servos, IMU, pressure / depth sensing, and stereo cameras into a free-swimming prototype with end-to-end control, communications, Steam Deck operation, logging, and safety.
- Built a **50 Hz supervisory loop and 200 Hz dual-motor position-command pipeline** over CAN / MIT for two AK45-10 actuators; achieved synchronized flapping up to 7 Hz, differential steering, phase resynchronization, cruise / manual override, and smooth return-to-neutral.
- Calibrated the stereo rig and implemented StereoSGBM depth; developed a Steam Deck console for gamepad commands, state feedback, stereo / depth display, E-Stop, and record / replay; validated stable teleoperation at **3 m depth**.
- Implemented layered safety with control-link watchdogs, link-loss return-to-neutral, actuator enable gating, unified E-Stop / shutdown paths, and staged thermal management.

Antagonistic VCA Joint | Wireless Control & Hall-Sensor Calibration

2026

SolidWorks · ESP32-C6 · Embedded C / C++ · PID · AS5600L · ESP-NOW · Python

- Designed the antagonistic dual-VCA joint and actuator / sensor layout in SolidWorks; built a Commander / Driver dual-ESP32-C6 architecture with local sensing / actuation on Driver and acknowledged ESP-NOW commands, queued reception, status feedback, and browser control through Commander.
- Implemented open-loop force adjustment and **500 Hz PID angle control**; demonstrated >30 W peak power per module, a linear PWM-to-force response, and closed-loop positioning over $\pm 45^\circ$.
- Added sensor filtering, dead-zone compensation, mechanical limits, encoder-fault latching, priority E-Stop, and automatic output shutdown on link loss.

CLD Soft Robotic Propulsor | Passive Impedance Shaping & Hydrodynamic Validation

2025–Present

IROS 2026 Accepted (First Author) · CLD · PIV · Experimental Design

- Independently designed and fabricated four coverage variants of a frequency-selective constrained-layer-damping (CLD) propulsor using a symmetric PLA / acrylic-foam / PET laminate.
- Built an air-to-water validation workflow: 1–5 Hz bending tests for complex impedance; water-tunnel propulsion at $St = 0.2$ –0.8 with thrust and phase-resolved PIV; and virtual-mass trials for free-swimming acceleration.
- At $St = 0.8$, increased thrust from **0.16 to 0.51 N (>200%)**, peak acceleration from 0.039 to 0.191 m/s² ($\sim 5\times$), and terminal speed from 0.105 to 0.291 m/s ($\sim 3\times$).

Teensy · TCA9548A · I2C · MS5803 · PyTorch

- Designed a five-channel pressure acquisition system and used Teensy hardware triggering to place **80 Hz pressure measurements and PIV on a shared time base**.
- Collected 54 synchronized flow / geometry cases for training and validation; built Python pipelines for cleaning, per-channel zero-offset correction, and pressure-PIV pairing.
- Designed dual-path attention networks for sparse PIV and pressure, vorticity-weighted losses, and ensemble-uncertainty active learning for full-field flow reconstruction.

SELECTED PUBLICATIONS

- **Feng, Q.**, Roberts, O. A., Zhong, Q. "Passive Phase-Oriented Impedance Shaping for Rapid Acceleration in Soft Robotic Swimmers." **IROS 2026, accepted (first author)**; arXiv:2603.03537.
- **Feng, Q.**, Han, T., Zhong, Q. "Lift Reversal from Vortex-Surface Phase Coupling in a Heaving Foil near a Free Surface." **Journal of Fluid Mechanics, under review (first author)**; arXiv:2512.12485.
- Zhang, Z., **Feng, Q.**, Zhu, Y., Zhong, Q. "Inertial Effects on the Mechanical Efficiency of a Semi-Passive Oscillating Hydrofoil Energy Harvester." **Renewable Energy, under review (second author)**; arXiv:2606.11126.
- Zhu, Y., Liu, L., Han, T., **Feng, Q.**, et al. "Wavenumber Affects the Lift of Ray-Inspired Fins near a Substrate." **J. R. Soc. Interface 22:20250276 (2025), published**.

HONORS & PRESENTATIONS

- **Link Foundation Ocean Engineering & Instrumentation Fellowship** | July 2024
- **APS Division of Fluid Dynamics Annual Meeting Presentation** | Nov. 2024
- **CoMFRE Annual Meeting Presentation** | Oct. 2024